Lesson 21: Nonparametric Inference

BSTA 551: Statistical Inference

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Lesson 21: Nonparametric Inference

Review: Lessons 18–20

- **Lesson 18:** Parametric two-sample tests — Welch t, pooled t, z-test for proportions, F-test
- **Lesson 19:** Paired t-test — reduces to one-sample inference on $D_i = X_{1i} - X_{2i}$
- **Lesson 20:** Bootstrap CIs (resample separately) and permutation tests (relabel pooled data)

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Today's Goals:

1. Understand what "nonparametric" means and when it is preferred over parametric tests
2. Derive and apply the **sign test** for paired data (DBC 14.1)
3. Derive and apply the **Wilcoxon signed-rank test** for paired data (DBC 14.2)
4. Derive and apply the **Wilcoxon rank-sum test** for two independent samples (DBC 14.3)
5. Understand **asymptotic relative efficiency (ARE)** and the power trade-offs it quantifies
6. Apply all three tests to medical examples and compare conclusions

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What “Nonparametric” Is NOT

Nonparametric $\neq$ assumption-free. These tests still require:

- **Independence** of observations (or independence of pairs)
- **Continuity** of the underlying distribution (for rank-based tests — ties deflate test power)
- **Symmetry** of D_i about 0 (for the Wilcoxon signed-rank test)

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Why does it matter in biomedical research?

- **Biomarkers** (CRP, troponin, ALT) are right-skewed with occasional extreme values
- **Ordinal scales** (pain VAS, Likert, disability scores) — arithmetic means may be misleading
- **Small** n — CLT convergence too slow to trust normality-based tests

Roadmap for Today

Test	Data Structure	H_0	Key Assumption	Reference
Sign test	Paired / one-sample	Median $D = 0$	None beyond independence	DBC 14.1
Wilcoxon signed-rank	Paired / one-sample	Median $D = 0$	Symmetric, continuous D_i	DBC 14.2
Wilcoxon rank-sum	Two independent samples	$F_1 = F_2$	Continuous distributions	DBC 14.3

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Information used by each test:

Test	Uses direction of D_i?	Uses magnitude of D_i?
Sign test	✓	✗
Wilcoxon signed-rank	✓	✓ (via ranks)
Paired t-test	✓	✓ (exact values)

Each test makes a different assumption about the data in exchange for the information it extracts.

The Sign Test (DBC 14.1)

Sign Test: Setup and Motivation

Setting: Paired differences $D_i = X_{1i} - X_{2i}, i = 1, \ldots, n$.

Goal: Test whether the **population median** of the D_i equals zero, without assuming any particular shape.

Key insight: If the true median of D_i is zero, then $P(D_i > 0) = P(D_i < 0) = 1/2$ (for continuous distributions).

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Test statistic: Count the positive differences.

$$S^+ = \#\{i : D_i > 0\}$$

Ties ($D_i = 0$): excluded. Let $n^* = \#\{i : D_i \neq 0\}$.

Null distribution: Under $H_0 : P(D_i > 0) = 1/2$,

$$\boxed{S^+ \sim \text{Binomial}(n^*, 1/2)}$$

This follows because each non-zero D_i is independently positive or negative with equal probability under H_0.

Sign Test: Null Distribution Derivation

Why exactly $\text{Binomial}(n^*, 1/2)$?

Define $Z_i = \mathbf{1}(D_i > 0)$ for each i with $D_i \neq 0$. Under H_0:

- $P(Z_i = 1) = P(D_i > 0) = 1/2$ (median is 0, distribution continuous)
- $P(Z_i = 0) = P(D_i < 0) = 1/2$
- $Z_1, \ldots, Z_{n^*}$ are **independent** (pairs are independent)

Therefore $Z_i \overset{iid}{\sim} \text{Bernoulli}(1/2)$ and

$$S^+ = \sum_{i=1}^{n^*} Z_i \sim \text{Binomial}(n^*, 1/2)$$

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Therefore $Z_i \overset{iid}{\sim} \text{Bernoulli}(1/2)$ and

$$S^+ = \sum_{i=1}^{n^*} Z_i \sim \text{Binomial}(n^*, 1/2)$$

This is exact — no asymptotic approximation needed. The binomial distribution is completely tractable.

Large-sample approximation: For large n^*, by the CLT:

$$Z_S = \frac{S^+ - n^*/2}{\sqrt{n^*/4}} \dot{\sim} N(0, 1)$$

Sign Test: P-values

Alternative	Exact p-value	Interpretation
$H_a\colon \tilde{\mu}_D > 0$	$P(S^+ \geq s^+)$	More positives than expected
$H_a\colon \tilde{\mu}_D < 0$	$P(S^+ \leq s^+)$	Fewer positives than expected
$H_a\colon \tilde{\mu}_D \neq 0$	$2\min\big(P(S^+ \geq s^+),\ P(S^+ \leq s^+)\big)$	Either tail

All probabilities from $\mathrm{Bin}(n^*, 1/2)$ — exact, no normality required.

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Confidence Interval for the Median (Sign Test)

An exact CI for the **population median** is based on order statistics. The $100(1-\alpha)\%$ CI for the median is $(D_{(j)}, D_{(n^*-j+1)})$ where j is chosen so $P(\mathrm{Bin}(n^*, 1/2) \leq j-1) \leq \alpha/2$.

In R: `wilcox.test(d, conf.int = TRUE)` gives the Wilcoxon CI (preferred when symmetry holds). For the sign-based CI: use `SignifReg::sign.ci()` or compute directly from the binomial.

Sign Test: Medical Example (Pain and Physical Therapy)

Study: 15 patients rate pain on a 0–10 VAS scale **before** and **after** a 4-week physical therapy program.

Hypotheses: H_0: median change $= 0$ vs. H_a: median change < 0 (pain **decreases**; $D_i = \text{after} - \text{before}$).

```r
pain_before <- c(8, 7, 9, 6, 8, 7, 9, 5, 8, 7, 6, 9, 8, 7, 6)
pain_after  <- c(5, 6, 6, 4, 5, 3, 7, 3, 5, 4, 4, 6, 5, 4, 3)
d_pain      <- pain_after - pain_before

cat("D = after - before:\n", d_pain, "\n\n")
```

```
D = after - before:
 -3 -1 -3 -2 -3 -4 -2 -2 -3 -3 -2 -3 -3 -3 -3
```

```r
n_star  <- sum(d_pain != 0)
s_plus  <- sum(d_pain > 0)

cat(sprintf("Non-zero differences: n* = %d\n", n_star))
```

```
Non-zero differences: n* = 15
```

```r
cat(sprintf("Positive differences: S+ = %d\n", s_plus))
```

```
Positive differences: S+ = 0
```

```r
cat(sprintf("Negative differences: %d\n\n", sum(d_pain < 0)))
```

```
Negative differences: 15
```

```r
# One-sided p-value: H_a is median < 0 → few positives supports H_a
p_sign_one <- pbinom(s_plus, size = n_star, prob = 0.5)
```

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D = after - before:
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Non-zero differences: n* = 15
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cat(sprintf("Positive differences: S+ = %d\n", s_plus))
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```
Positive differences: S+ = 0
```

```r
cat(sprintf("Negative differences: %d\n\n", sum(d_pain < 0)))
```

```
Negative differences: 15
```

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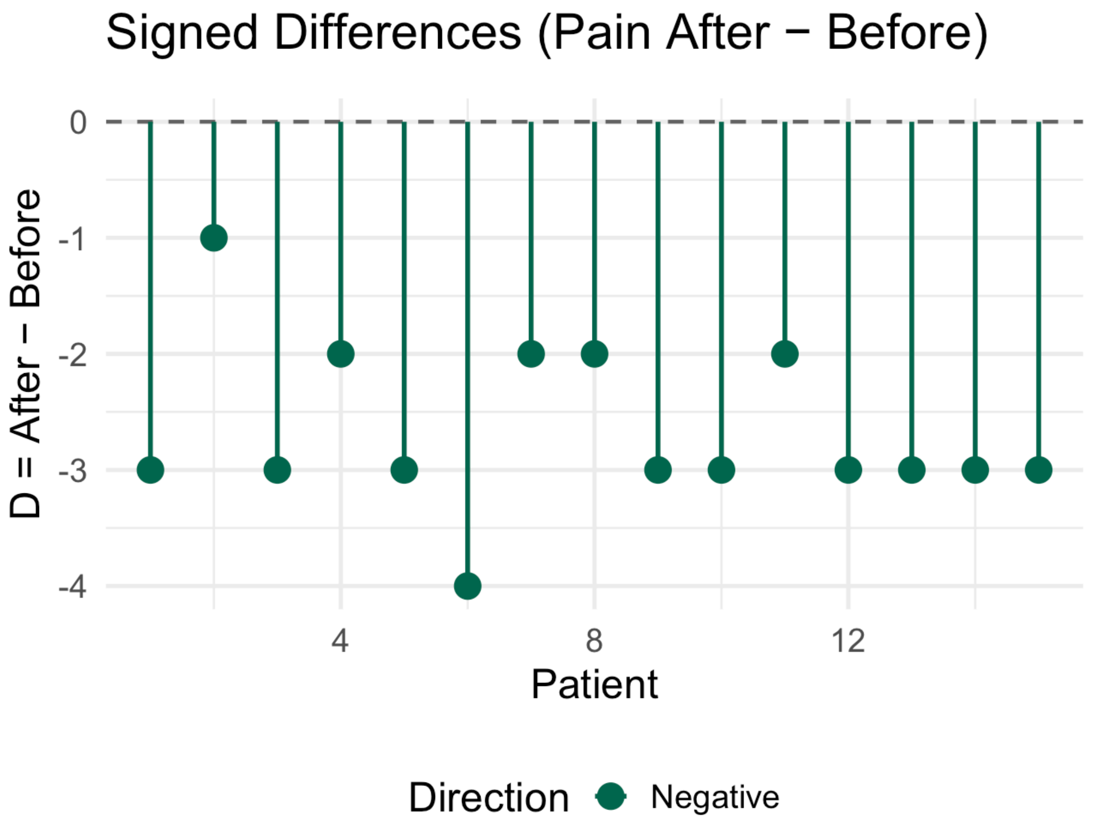
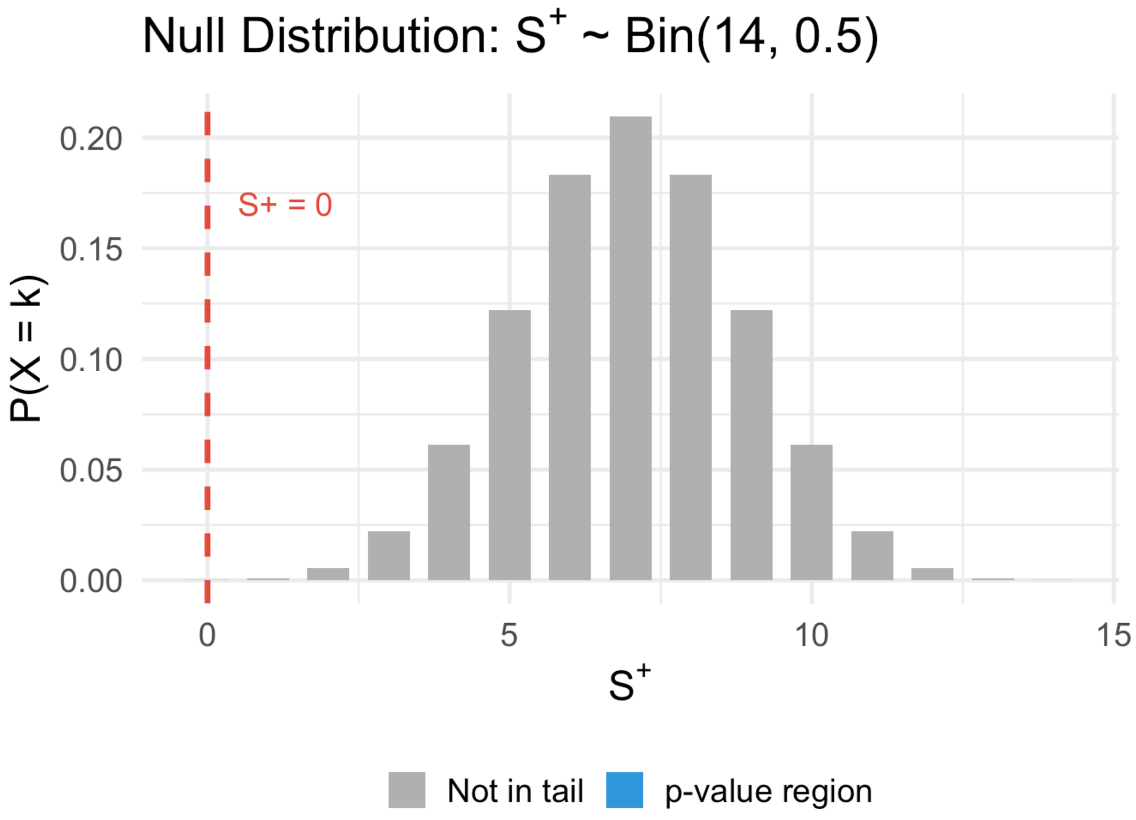
Sign Test: Visualizing the Evidence

▶ Code





Sign Test: Strengths and Limitations

Strengths:

- **No distributional assumptions** beyond independence — valid for any continuous or ordinal D_i
- Works when differences are measured on an **ordinal scale** (e.g., pain VAS, Likert), where arithmetic means are questionable
- The simplest nonparametric test to derive, compute, and explain

Limitations:

- Only uses the **sign** (direction) of each difference — discards all magnitude information
- **Lowest power** among the three paired tests when differences are continuous and symmetric
- Wide confidence intervals for the median, especially at small n

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> **When to Prefer the Sign Test**
>
> - Data are **ordinal**: arithmetic differences aren't on an interval scale
> - Differences are **highly asymmetric** with extreme outliers: sign is reliable, magnitude is not
> - You need the most **assumption-free** result and are willing to sacrifice power
> - Only the **direction** of each difference is observable (e.g., preferences without magnitudes)

Wilcoxon Signed-Rank Test (DBC 14.2)

Wilcoxon Signed-Rank: Motivation

The sign test throws away the **size** of each difference. Can we do better while still avoiding normality?

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Key idea: Replace the raw differences with their **ranks** by absolute value.

- Ranks preserve the *relative ordering* of the magnitudes
- They are less sensitive to extreme outliers than raw values
- If the distribution is symmetric about 0, the ranks of positive and negative differences should look "balanced" under H_0

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Key assumption added over sign test: The distribution of D_i is **symmetric** about its median.

- Under H_0, symmetric about 0
- Under H_a, symmetric about $\theta \neq 0$ (a location shift)

This rules out heavily skewed D_i – but covers most bell-shaped and even uniform-like distributions.

Wilcoxon Signed-Rank: Algorithm

Setup: $D_1, \ldots, D_n \overset{iid}{\sim}$ a continuous, symmetric distribution. H_0: median $= 0$.

Step-by-step:

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4. **Also define**: $W^- = \sum_{\{i:\, D_i < 0\}} R_i = \frac{n^*(n^*+1)}{2} - W^+$

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Under H_0 (symmetric about 0):

$$E[W^+] = \frac{n^*(n^*+1)}{4}, \qquad \mathrm{Var}(W^+) = \frac{n^*(n^*+1)(2n^*+1)}{24}$$

Large n^*: $\quad Z = \dfrac{W^+ - E[W^+]}{\sqrt{\mathrm{Var}(W^+)}} \,\dot{\sim}\, N(0,1)$

Why Does $E[W^+] = \frac{n^*(n^*+1)}{4}$?

Under H_0 and symmetry, each rank R_i is equally likely to be assigned to a positive or negative difference. Formally, let $\phi_i = \mathbf{1}(D_i > 0)$. Under H_0:

- $\phi_i \overset{iid}{\sim} \text{Bernoulli}(1/2)$, independent of the ranks R_i
- $W^+ = \sum_{i=1}^{n^*} \phi_i \cdot R_i$

Then:

$$E[W^+] = \sum_{i=1}^{n^*} E[\phi_i] \cdot R_i = \frac{1}{2} \sum_{i=1}^{n^*} R_i = \frac{1}{2} \cdot \frac{n^*(n^*+1)}{2} = \frac{n^*(n^*+1)}{4}$$

The variance follows from $\text{Var}(\phi_i) = 1/4$ and independence:

$$\text{Var}(W^+) = \sum_{i=1}^{n^*} R_i^2 \cdot \text{Var}(\phi_i) = \frac{1}{4} \sum_{i=1}^{n^*} i^2 = \frac{1}{4} \cdot \frac{n^*(n^*+1)(2n^*+1)}{6} = \frac{n^*(n^*+1)(2n^*+1)}{24}$$

Wilcoxon Signed-Rank: Continuity Correction and Ties

Continuity correction for the normal approximation:

$$Z = \frac{W^+ - E[W^+] \pm 0.5}{\sqrt{\mathrm{Var}(W^+)}}$$

Use -0.5 when $W^+ > E[W^+]$ (upper-tail test), $+0.5$ when $W^+ < E[W^+]$ (lower-tail test). R does this automatically.

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Correction for ties in ranking:

When there are many ties among the $|D_i|$, the variance formula understates the true variance. The corrected formula is:

$$\mathrm{Var}_{\text{ties}}(W^+) = \frac{1}{24}\left[n^*(n^*+1)(2n^*+1) - \frac{1}{2}\sum_j t_j(t_j-1)(t_j+1)\right]$$

where t_j is the size of the j-th group of tied absolute differences.

- When there are no ties, this reduces to the standard formula
- Many ties $\Rightarrow$ reduced variance $\Rightarrow$ more conservative test

Wilcoxon Signed-Rank: Medical Example

Study: 16 patients with Stage 1 hypertension. Systolic blood pressure (SBP, mmHg) measured before and 8 weeks after starting a new antihypertensive drug.

Hypotheses: H_0: median SBP change $= 0$ vs. H_a: median SBP change < 0 (drug lowers BP; $D_i = \text{after} - \text{before}$).

```r
sbp_before <- c(158, 142, 168, 135, 172, 145, 163, 138, 155, 149, 167, 141, 160, 153, 144,
sbp_after  <- c(145, 148, 152, 131, 158, 139, 170, 135, 142, 143, 151, 150, 148, 140, 149,
d_sbp      <- sbp_after - sbp_before

cat("D = after - before (mmHg):\n")
```

```
D = after - before (mmHg):
```

```r
cat(d_sbp, "\n\n")
```

```
-13 6 -16 -4 -14 -6 7 -3 -13 -6 -16 9 -12 -13 5 -15
```

```r
cat(sprintf("n = %d pairs | Positive (BP increased): %d | Negative (BP decreased): %d | Zer
            length(d_sbp), sum(d_sbp > 0), sum(d_sbp < 0), sum(d_sbp == 0)))
```

```
n = 16 pairs | Positive (BP increased): 4 | Negative (BP decreased): 12 | Zero: 0
```

```r
cat(sprintf("Mean change: %.1f mmHg | Median change: %.1f mmHg\n",
            mean(d_sbp), median(d_sbp)))
```

```
Mean change: -6.5 mmHg | Median change: -9.0 mmHg
```

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d_sbp      <- sbp_after - sbp_before

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```

```
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```

Most patients improved (SBP decreased), but **4 patients had higher SBP** after treatment — the kind of individual

Wilcoxon Signed-Rank: Step-by-Step Calculation

```r
# Step 1: Remove zeros (none here)
d_nz   <- d_sbp[d_sbp != 0]
n_star <- length(d_nz)

# Step 2: Rank absolute values (ties averaged automatically)
abs_ranks <- rank(abs(d_nz))

# Step 3: Compute W+
W_plus  <- sum(abs_ranks[d_nz > 0])
W_minus <- sum(abs_ranks[d_nz < 0])

cat(sprintf("n* = %d\n", n_star))
```

```
n* = 16
```

```r
cat(sprintf("W+ = %.1f   (sum of ranks for %d positive differences)\n",
            W_plus, sum(d_nz > 0)))
```

```
W+ = 23.0   (sum of ranks for 4 positive differences)
```

```r
cat(sprintf("W- = %.1f   (sum of ranks for %d negative differences)\n",
            W_minus, sum(d_nz < 0)))
```

```
W- = 113.0   (sum of ranks for 12 negative differences)
```

```r
cat(sprintf("Check: W+ + W- = %.1f  (should equal n*(n*+1)/2 = %.1f)\n\n",
            W_plus + W_minus, n_star*(n_star+1)/2))
```

Wilcoxon Signed-Rank: R Output and Hodges-Lehmann Estimate

```r
# R uses exact distribution for small n, Hodges-Lehmann estimate
wsr_result <- wilcox.test(sbp_after, sbp_before,
                          paired      = TRUE,
                          alternative = "less",
                          conf.int    = TRUE)
wsr_result |> tidy() |>
  select(statistic, p.value, conf.low, conf.high, method, alternative) |>
  print()
```

```
# A tibble: 1 × 6
  statistic p.value conf.low conf.high method                       alternative
      <dbl>   <dbl>    <dbl>     <dbl> <chr>                        <chr>
1        23  0.0106     -Inf     -3.00 Wilcoxon signed rank test wi… less
```

Wilcoxon Signed-Rank: R Output and Hodges-Lehmann Estimate

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# A tibble: 1 × 6
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      <dbl>   <dbl>    <dbl>     <dbl> <chr>                         <chr>
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```

The Hodges-Lehmann Estimator

The **Hodges-Lehmann (HL) estimator** of location shift associated with the signed-rank test is:

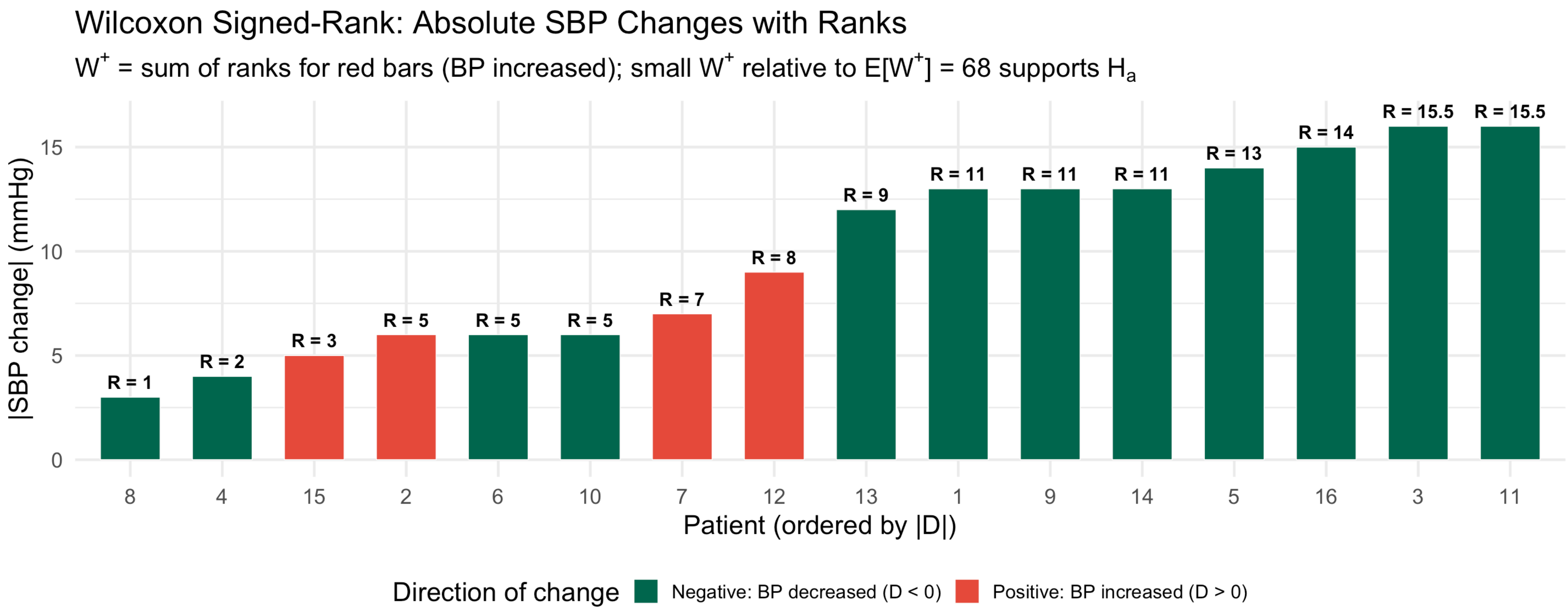
$$\hat{\theta}_{HL} = \text{median}\left\{\frac{D_i + D_j}{2} : 1 \leq i \leq j \leq n^*\right\}$$

the median of all **Walsh averages** (pairwise averages of differences, including self-averages).

- This is a natural point estimate when using the signed-rank test
- The confidence interval from `wilcox.test(..., conf.int = TRUE)` is the **Hodges-Lehmann CI**
- More robust to outliers than the sample mean, but still uses magnitude information

Visualizing the Rankings for the SBP Data

▶ Code



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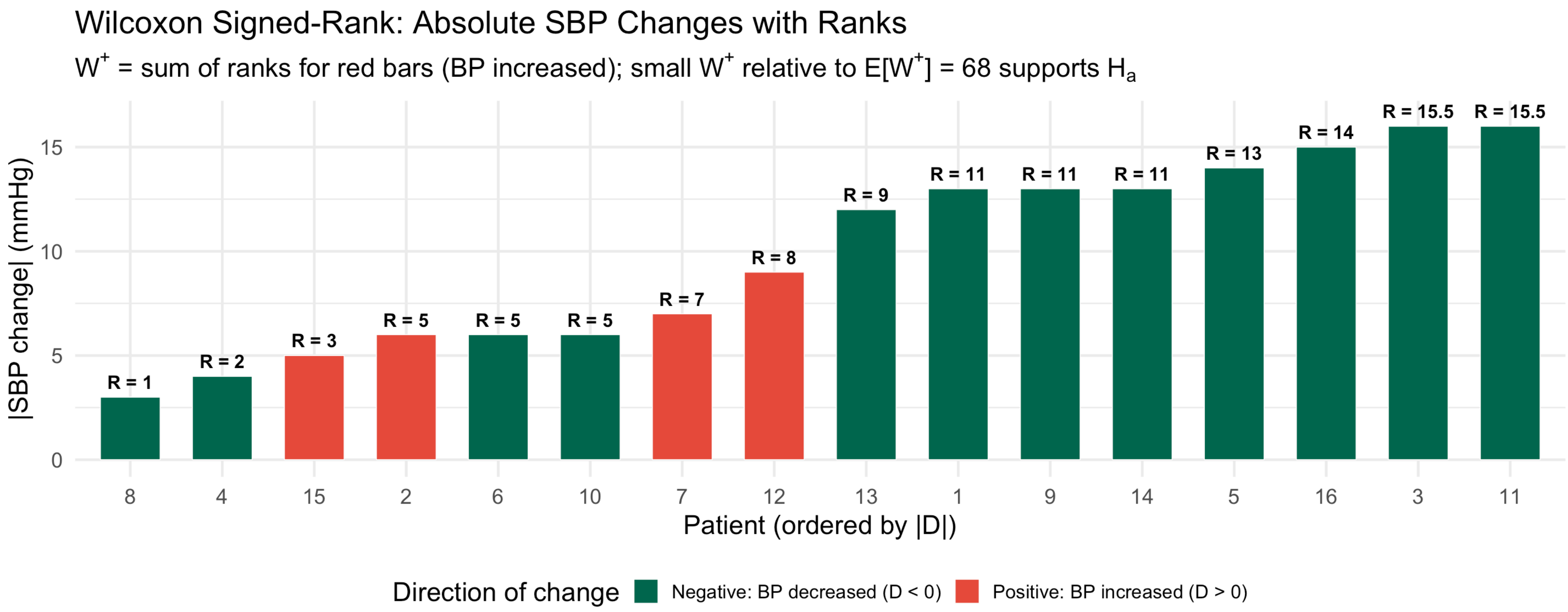
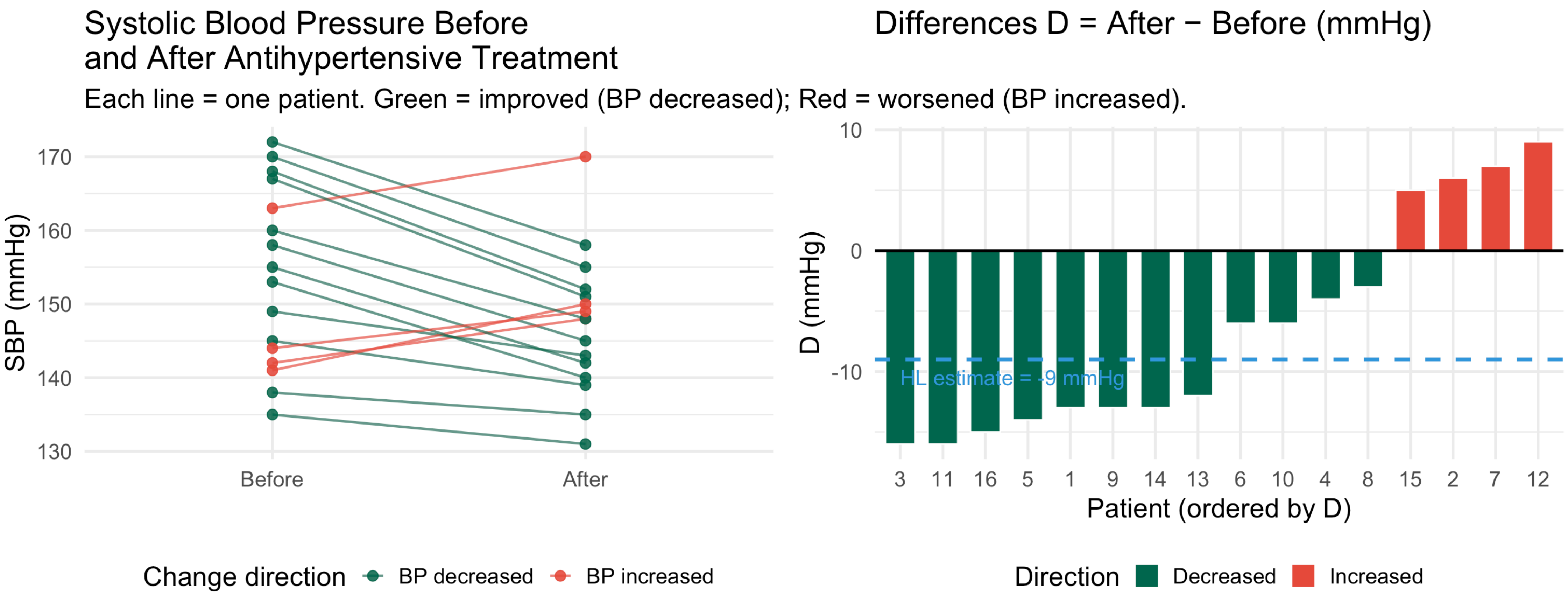


$W^+ = 23 \ll E[W^+] = 68$ – the positive-difference ranks are concentrated at low values, indicating the drug tends to reduce SBP.

Wilcoxon Signed-Rank: Visualizing the Paired Data

▶ Code



Comparing Sign Test, Wilcoxon Signed-Rank, and Paired *t* on SBP Data

```r
cat("=== SIGN TEST ===\n")
```

```
=== SIGN TEST ===
```

```r
n_s  <- sum(d_sbp != 0)
sp   <- sum(d_sbp > 0)
p_sign <- pbinom(sp, n_s, 0.5)
cat(sprintf("S+ = %d   n* = %d   one-sided p = %.5f\n\n", sp, n_s, p_sign))
```

```
S+ = 4   n* = 16   one-sided p = 0.03841
```

```r
cat("=== WILCOXON SIGNED-RANK ===\n")
```

```
=== WILCOXON SIGNED-RANK ===
```

```r
wilcox.test(sbp_after, sbp_before, paired = TRUE,
            alternative = "less", conf.int = TRUE) |>
  tidy() |>
  select(statistic, p.value, conf.low, conf.high) |>
  print()
```

```
# A tibble: 1 × 4
  statistic p.value conf.low conf.high
      <dbl>   <dbl>    <dbl>     <dbl>
1        23  0.0106     -Inf     -3.00
```

```r
cat("\n")
```

```r
cat("=== PAIRED t-TEST ===\n")
```

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```r
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```

Wilcoxon Rank-Sum Test (DBC 14.3)

Rank-Sum Test: Two Independent Samples

Setting: $X_{11}, \ldots, X_{1n_1} \overset{iid}{\sim} F_1$ and $X_{21}, \ldots, X_{2n_2} \overset{iid}{\sim} F_2$, independent samples.

H_0: $F_1 = F_2$ – the two samples come from the **same distribution**.

Under a **location-shift model** ($F_2(x) = F_1(x - \Delta)$), this is equivalent to $H_0 : \Delta = 0$.

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Algorithm:

1. Pool all $N = n_1 + n_2$ observations together; rank from 1 to N (average ties)
2. Compute $W = \sum_{\text{group 1}} R_i$ = sum of ranks for group 1 observations
3. Under H_0: **every allocation of ranks to groups is equally likely**

$$E[W] = \frac{n_1(N+1)}{2}, \qquad \text{Var}(W) = \frac{n_1 n_2(N+1)}{12}$$

For large samples: $Z = \dfrac{W - E[W]}{\sqrt{\text{Var}(W)}} \mathrel{\dot\sim} N(0,1)$

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The exact null distribution of W is computed by R for small n_1, n_2 (`wilcox.test` with `exact = TRUE`).

Rank-Sum Test: Why This Null Distribution?

Under $H_0 : F_1 = F_2$, the N pooled observations are exchangeable. The group labels (group 1 or group 2) are equally likely to be assigned to any set of n_1 positions out of N.

There are $\binom{N}{n_1}$ equally-likely allocations.

W = sum of group-1 ranks ranges from $\frac{n_1(n_1+1)}{2}$ (smallest n_1 ranks) to $\frac{n_1(2n_2+n_1+1)}{2}$ (largest n_1 ranks).

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Expected value:

$$E[W] = n_1 \cdot E[\text{rank of a single obs.}] = n_1 \cdot \frac{N+1}{2}$$

since under H_0, each rank from 1 to N is equally likely for any observation.

Variance:

$$\text{Var}(W) = \frac{n_1 n_2 (N+1)}{12}$$

derived from the variance of sampling without replacement from $\{1, \ldots, N\}$.

The Mann–Whitney U Statistic

The **Mann-Whitney** U is equivalent to W and has a useful probabilistic interpretation:

$$U = W - \frac{n_1(n_1+1)}{2}$$

U counts the number of concordant pairs:

$$U = \#\{(i,j) : X_{1i} > X_{2j}\}$$

out of all $n_1 \times n_2$ pairs. Under H_0: $E[U] = n_1 n_2/2$.

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$U/(n_1 n_2)$ is a consistent estimator of $P(X_1 > X_2)$:

- $P(X_1 > X_2) = 1/2$ under H_0
- $P(X_1 > X_2) > 1/2$ when group 1 tends to be larger
- This is the **area under the ROC curve (AUC)** — a key measure in diagnostic medicine

> **Mann-Whitney U and the ROC Curve**
>
> If you use group membership as a "predictor" and a continuous biomarker as the "score," the Mann-Whitney U statistic normalized by $n_1 n_2$ equals the AUC of the ROC curve. This connects nonparametric testing directly to classification performance.

Rank-Sum Test: Ties Correction

When there are ties among the pooled observations, the variance is corrected:

$$\mathrm{Var}_{\text{ties}}(W) = \frac{n_1 n_2}{12}\left[N + 1 - \frac{\sum_j t_j(t_j^2 - 1)}{N(N-1)}\right]$$

where t_j = size of the j-th tied group in the combined sample.

- Ties reduce the variance (less information in the ranks)
- The correction makes the Z-statistic slightly larger in absolute value
- R applies this automatically in `wilcox.test()`

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Hodges-Lehmann Estimator for Two Samples:

$$\hat{\Delta}_{HL} = \text{median}\{X_{1i} - X_{2j} : i = 1, \ldots, n_1;\ j = 1, \ldots, n_2\}$$

The median of all $n_1 \times n_2$ pairwise differences. This is the natural point estimate of the location shift Δ under the rank-sum framework.

Rank-Sum Test: Medical Example (Serum CRP)

Study: Compare serum CRP (mg/L), a right-skewed inflammatory biomarker, between patients **with** and **without** post-operative infections.

```r
set.seed(551)
crp_infect  <- rgamma(18, shape = 2.0, rate = 0.08)  # mean ~25 mg/L
crp_control <- rgamma(20, shape = 1.2, rate = 0.12)  # mean ~10 mg/L

cat(sprintf("Infection:  n = %d   mean = %.1f   median = %.1f   sd = %.1f   range = [%.1f,
            length(crp_infect),  mean(crp_infect),
            median(crp_infect),  sd(crp_infect),
            min(crp_infect), max(crp_infect)))
```

```
Infection:  n = 18   mean = 19.9   median = 22.5   sd = 10.1   range = [3.5, 33.8]
```

```r
cat(sprintf("Control:     n = %d   mean = %.1f   median = %.1f   sd = %.1f   range = [%.1f,
            length(crp_control), mean(crp_control),
            median(crp_control), sd(crp_control),
            min(crp_control), max(crp_control)))
```

```
Control:     n = 20   mean = 11.7   median = 6.4   sd = 10.3   range = [1.7, 36.3]
```

```r
N  <- length(crp_infect) + length(crp_control)
n1 <- length(crp_infect)
n2 <- length(crp_control)

W_obs  <- sum(rank(c(crp_infect, crp_control))[1:n1])
```

Rank-Sum Test: Comparison with Welch *t*

```
cat("--- Welch t-test ---\n")
```

```
--- Welch t-test ---
```

```
t.test(crp_infect, crp_control, alternative = "greater") |>
  tidy() |>
  select(statistic, parameter, p.value, conf.low, method) |>
  print()
```

```
# A tibble: 1 × 5
  statistic parameter p.value conf.low method
      <dbl>     <dbl>   <dbl>    <dbl> <chr>
1      2.49      35.7 0.00876     2.67 Welch Two Sample t-test
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```
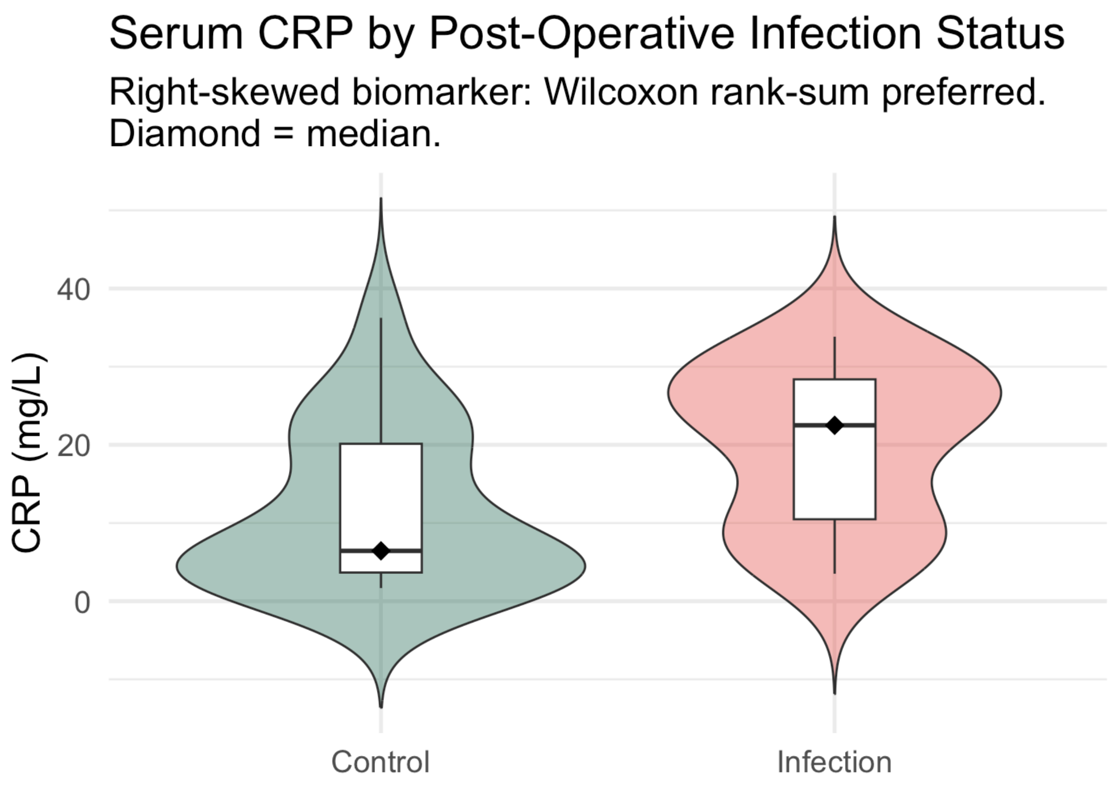
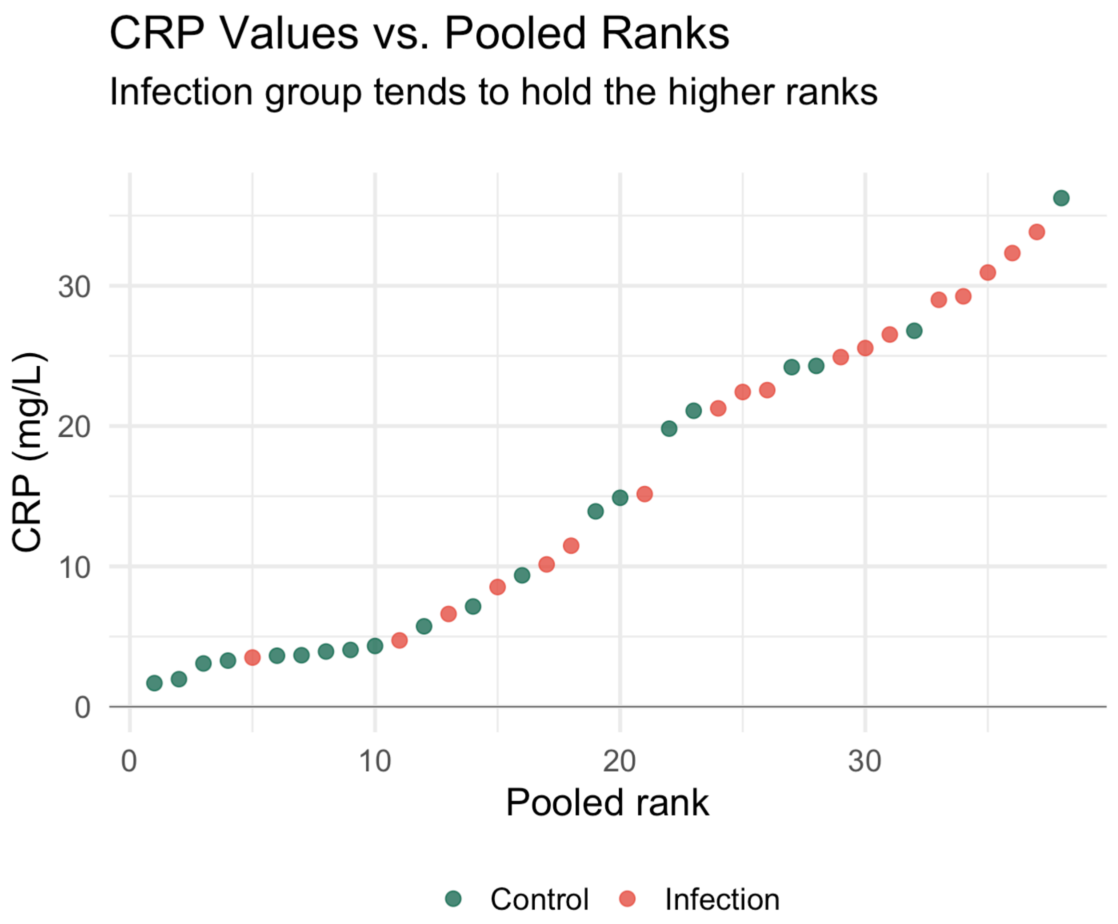

Both tests reject H_0, but note:

- CRP is strongly **right-skewed** with outliers → t-test p-value is less reliable
- Wilcoxon rank-sum p-value is based on a **distribution-free** null
- Hodges-Lehmann estimate of location shift: `conf.low` from `wilcox.test`
- With small n and skewed biomarkers, the rank-sum test is the **preferred primary analysis**

Visualizing the CRP Data

▶ Code





Asymptotic Relative Efficiency

What Is Asymptotic Relative Efficiency?

Definition: The ARE of test T_1 relative to test T_2 is:

$$\text{ARE}(T_1, T_2) = \lim_{n \to \infty} \frac{n_2^*}{n_1^*}$$

where n_i^* = sample size needed by test T_i to achieve the **same power** at the same significance level α and the same effect size.

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Interpretation:

- $\text{ARE}(T_1, T_2) = 0.95$: T_1 needs $n_2/0.95 \approx 1.05n_2$ observations – 5% more
- $\text{ARE}(T_1, T_2) = 1.50$: T_1 needs only $n_2/1.50 \approx 0.67n_2$ – 33% fewer!
- $\text{ARE}(T_1, T_2) = 0.50$: T_1 needs twice as many observations

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- $\text{ARE}(T_1, T_2) = 0.50$: T_1 needs twice as many observations

ARE is computed under a specific distributional family (the **Pitman local alternatives** framework) and tells us about asymptotic power, not finite-sample behavior — but it is a reliable guide in practice.

ARE: Pitman Efficiency Results

The Pitman ARE for location-shift alternatives $H_a : \theta = \theta_n = c/\sqrt{n}$ (DBC 14.5, DBC p. 681):

Distribution of D_i	**ARE(sign / t)**	**ARE(signed-rank / t)**
Normal	$2/\pi \approx 0.637$	$3/\pi \approx 0.955$
Logistic	$\pi^2/12 \approx 0.822$	$\pi^2/9 \approx 1.097$
Double exponential (Laplace)	2.000	$3/2 = 1.500$
Uniform	$1/3 \approx 0.333$	1.000
Cauchy (very heavy tails)	∞	∞
t_ν (ν small)	> 1	> 1

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The Remarkable Result for the Signed-Rank Test

- At **normality**: ARE(signed-rank, t) = $3/\pi \approx 0.955$ — only **4.5% efficiency loss**
- The signed-rank test is **never catastrophically bad** — for all symmetric distributions, ARE ≥ 0.864 (Hodges-Lehmann lower bound)
- For heavy-tailed or non-normal distributions, the signed-rank test can be **substantially more powerful**

This makes it an excellent “default” for small n when normality is uncertain.

Where Do the ARE Formulas Come From?

For the **signed-rank test** vs. the t-test under the normal distribution:

The ARE can be computed using the **Pitman formula**:

$$\text{ARE}(W^+, t) = \frac{\left[E\left\{f(D)\right\}\right]^2 \cdot \text{Var}(D)}{\left[\frac{1}{2}f(0)\right]^2}$$

where f is the density of D_i.

For $D_i \sim N(0, \sigma^2)$:

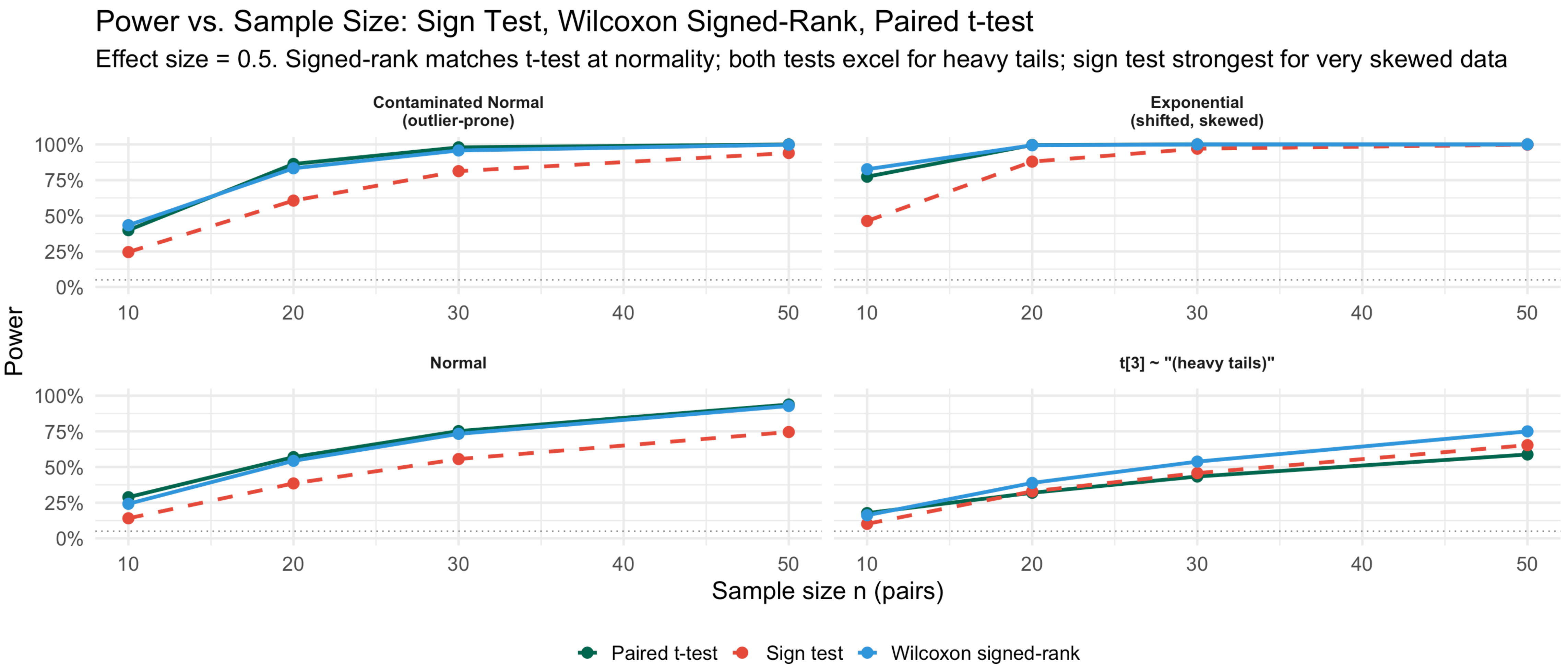
- $f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-x^2/(2\sigma^2)}$, so $f(0) = \frac{1}{\sigma\sqrt{2\pi}}$
- $E\{f(D)\} = \int_{-\infty}^{\infty} f(x)^2\,dx = \frac{1}{2\sigma\sqrt{\pi}}$

This gives $\text{ARE} = \frac{3}{\pi} \approx 0.955$.

The analogous result for the sign test is $\text{ARE}(\text{sign}, t) = 4\sigma^2 f(0)^2 = 2/\pi$ under normality.

ARE: Power Simulation Across Distributions

▶ Code



ARE: Interpreting the Simulation

The simulation confirms the Pitman ARE theory:

- **Normal distribution**: Signed-rank $\approx t$-test power (ARE ≈ 0.955). Sign test noticeably weaker (ARE ≈ 0.637).
- t_3 **distribution (heavy tails)**: Both nonparametric tests **outperform** the t-test. Signed-rank especially strong.
- **Exponential (skewed)**: Signed-rank invalid in principle (asymmetric D_i), but often still competitive in practice. Sign test surprisingly strong.
- **Contaminated normal**: Even modest outlier contamination makes nonparametric tests superior.

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- **Exponential (skewed)**: Signed-rank invalid in principle (asymmetric D_i), but often still competitive in practice. Sign test surprisingly strong.
- **Contaminated normal**: Even modest outlier contamination makes nonparametric tests superior.

Practical Takeaway for Biomedical Research

Many biomarkers (CRP, troponin, liver enzymes) follow right-skewed, heavy-tailed distributions.

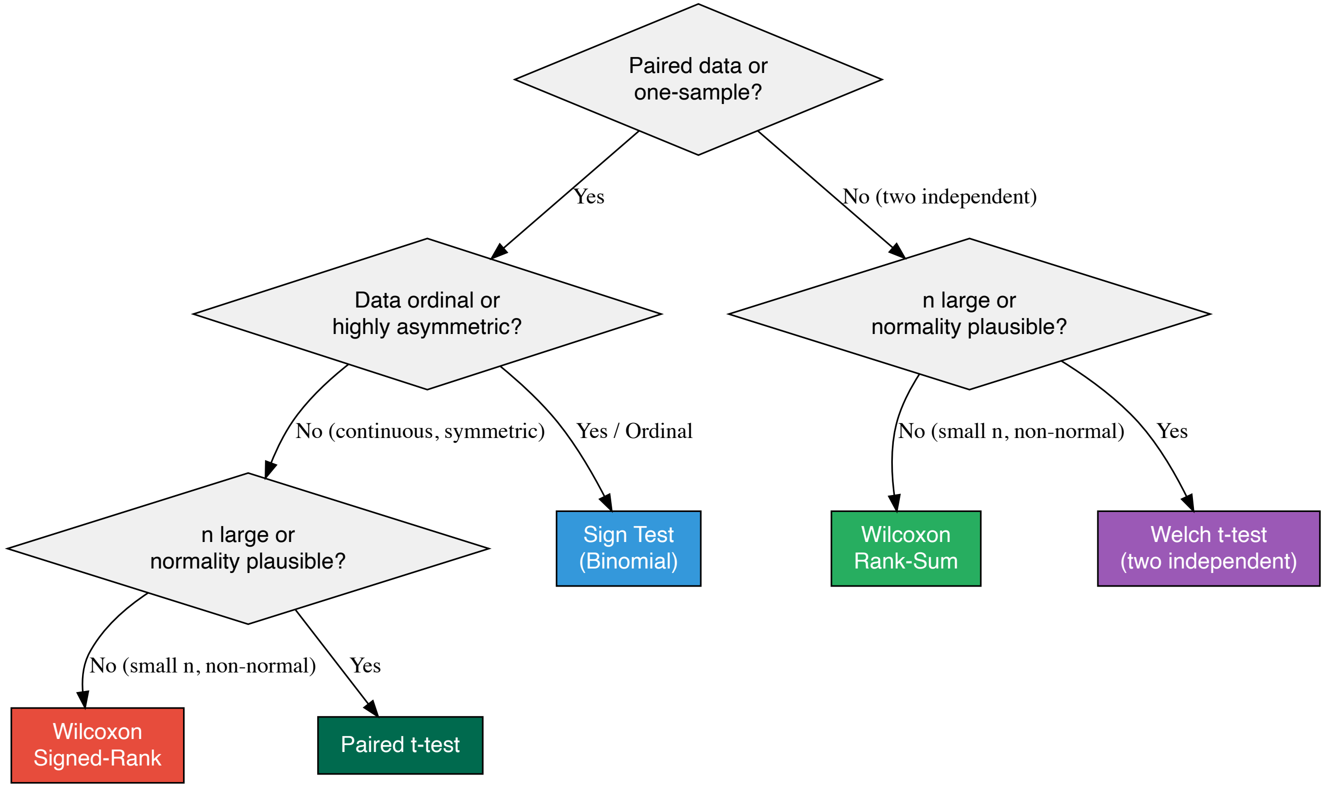
For such data with small n:

- **Sign test**: if ordinal data or very extreme outliers
- **Wilcoxon signed-rank**: best default for continuous paired data with uncertain normality
- **Wilcoxon rank-sum**: best default for independent two-group comparisons

Comparing All Three Tests

Decision Flowchart: Which Test?

▶ Code



Side-by-Side on the Pain Score Data (Sign Test Example)

```r
cat("=== SIGN TEST ===\n")
```

```
=== SIGN TEST ===
```

```r
n_s <- sum(d_pain != 0); sp <- sum(d_pain > 0)
cat(sprintf("S+ = %d    n* = %d    one-sided p = %.5f\n\n",
            sp, n_s, pbinom(sp, n_s, 0.5)))
```

```
S+ = 0    n* = 15    one-sided p = 0.00003
```

```r
cat("=== WILCOXON SIGNED-RANK ===\n")
```

```
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```r
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  tidy() |>
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  print()
```

```
# A tibble: 1 × 5
  statistic  p.value conf.low conf.high method
      <dbl>    <dbl>    <dbl>     <dbl> <chr>
1         0 0.000259     -Inf     -2.50 Wilcoxon signed rank test with continui…
```

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Summary Table: When to Use Each Test

Test	Use When	Caution
Sign test	Ordinal data; only sign matters; extreme asymmetry of D_i	Wastes magnitude information; lowest power
Wilcoxon signed-rank	Continuous paired data; small n; uncertain normality	Requires **symmetric** D_i; not for ordinal-only data
Wilcoxon rank-sum	Two independent continuous groups; small n; skewed	Requires **continuous** distributions; exact p-value affected by ties
Paired t-test	Paired; approximately normal D_i or $n \geq 30$	Sensitive to heavy tails and outliers for small n
Welch t-test	Two independent groups; approximately normal or $n_i \geq 30$	Heavy tails inflate Type I error for small n

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Default Recommendation for Biomedical Research

For **paired** continuous D_i with $n < 30$ and uncertain normality: use **Wilcoxon signed-rank**. For **two independent** groups with small n and skewed biomarkers: use **Wilcoxon rank-sum**.

If assumptions are met, the parametric test gains only ~5% efficiency — a small price for robustness.

Key Takeaways

Key Takeaways (DBC 14.1–14.3)

Nonparametric Inference: Main Points

1. **Sign test:** $S^+ \sim \text{Bin}(n^*, 1/2)$ under H_0; uses only direction of D_i; valid for ordinal data; no symmetry assumption; lowest power of the three
2. **Wilcoxon signed-rank:** ranks $|D_i|$, sums ranks of positive differences; requires symmetric (not necessarily normal) continuous D_i; ARE vs. t-test $= 3/\pi \approx 0.955$ at normality — minimal efficiency loss, with gains for non-normal data; Hodges-Lehmann estimator is the associated point estimate
3. **Wilcoxon rank-sum (Mann–Whitney U):** pools and ranks all N observations; sums group-1 ranks; tests $H_0 : F_1 = F_2; U/(n_1 n_2)$ estimates $P(X_1 > X_2)$ = AUC; Hodges-Lehmann estimate = median of all $n_1 \times n_2$ pairwise differences
4. **ARE quantifies the asymptotic power trade-off:** signed-rank ARE ≥ 0.864 vs. t-test for all symmetric distributions (Hodges-Lehmann lower bound) — never much worse at normality, potentially much better for heavy tails
5. **Nonparametric $\neq$ assumption-free:** signed-rank requires **symmetric** D_i; rank-sum requires **continuous** distributions; both require **independent** observations; ties require variance corrections
6. **Variance corrections for ties:** both tests have modified variance formulas; R applies them automatically; heavy ties (e.g., ordinal data) reduce precision of rank-based tests — consider the sign test or permutation tests

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Questions?

Thank you!